

Empirical Proof Record: VALIDATED

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1. Claim

The posterior 94% HDI width on μ_Φ contracts at the theoretical $\sqrt{N}$ rate as the number of observations grows. Specifically, $\text{HDI_width}(N=200) / \text{HDI_width}(N=20) \leq 0.40$ (theoretical contraction factor $\sqrt{(20/200)} = 0.316$; the 0.40 ceiling allows finite-sample slack).

- **Operationalisation:** Fit a PyMC NUTS posterior to subsamples of the Φ trajectory at sizes 20, 50, 100, 200. Posterior model: $\mu \sim \text{Normal}(\text{prior})$, $\sigma \sim \text{HalfNormal}$, $y_i \sim \text{Normal}(\mu, \sigma)$. Compute the 94% HDI width on μ at each size; the contraction metric is $\text{HDI_width}(N=200) / \text{HDI_width}(N=20)$.
- **Threshold:** `hdi_contraction_ratio <= 0.4 ratio`
- **H0:** `hdi_contraction_ratio > 0.40` (posterior fails to tighten at the predicted rate — data may be non-Gaussian, non-stationary, or sampler miscalibrated)
- **H1:** `hdi_contraction_ratio <= 0.40` (posterior tightens approximately at the theoretical $\sqrt{N}$ rate, confirming Bayesian inference behaves correctly on the Φ trajectory)

2. Verdict

VALIDATED — observed 0.27735050659795846

HDI width $N=20 \rightarrow N=200$: 0.0654 $\rightarrow$ 0.0181; contraction ratio 0.2774 (theoretical 0.3162; ceiling 0.4000). Empirical Φ mean=0.6140 sd=0.0680 (n=200).

3. Evidence

Pillar	Statistic	Value	95% CI	Library
pymc_nuts_posterior	hdi_contraction_ratio	0.27735050659795846	—	pymc 6.0.0

4. Pre-registration

- Registered at: 2026-05-31T18:33:45.989519+00:00
- Config hash: a47e4fd070e7935308f57886af093806...
- Data hash: 9ef317fe291445319fcba522d29836dc...

5. Signature

7c7104bffc6c134b4ebabd1750551aa7c617ab5ba8b383b757b9a30d9d476e34